

EXAM 7 – SPRING 2014

21. (5 points)

Given the following data:

<u>Class</u>	<u>Frequency</u>	<u>Severity</u>
1	0.20	\$15,000
2	0.05	25,000
3	0.40	5,000
4	0.25	3,000
5	0.10	12,000

a. (1.5 points)

Calculate the Pearson product-moment correlation coefficient between the frequency and severity.

b. (1.5 points)

Calculate the Kendall τ statistic between the frequency and severity.

c. (1 point)

Identify and briefly describe two potential weaknesses of the Pearson product-moment correlation coefficient as a measure of dependency.

d. (0.5 point)

Briefly describe whether the use of the Kendall τ statistic would address the weaknesses identified in part c. above.

e. (0.5 point)

Briefly describe one advantage and one disadvantage of using a scatterplot to evaluate dependency instead of using the Kendall τ statistic or the Pearson product-moment correlation coefficient.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 21

TOTAL POINT VALUE: 5

LEARNING OBJECTIVE: C2: Describe the use of enterprise-wide risk modeling and aggregation techniques; and C3: Evaluate and select appropriate models to handle diverse risks, including stochastic approaches

SAMPLE ANSWERS

Part a: 1.5 points

Sample Answer 1

$$\rho = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \times \sum (Y_i - \bar{Y})^2}}$$

Frequency	Severity	$X_i - \bar{X}$	$Y_i - \bar{Y}$
0.2	15,000	0.00	3,000
0.05	25,000	-0.15	13,000
0.4	5,000	0.20	-7,000
0.25	3,000	0.05	-9,000
0.1	12,000	-0.10	0

$$\begin{array}{rcl} \bar{X} & & 0.2 \\ \bar{Y} & & 12,000 \\ \sqrt{\sum (X_i - \bar{X})^2} & & 0.0750 \\ \sqrt{\sum (Y_i - \bar{Y})^2} & & 308,000,000 \\ \sum (X_i - \bar{X})(Y_i - \bar{Y}) & & -3,800 \end{array}$$

$$= \frac{-3,800}{\sqrt{0.0750 \times 308000000}} = -79\%$$

Sample Answer 2

$$\rho = \frac{\overline{XY} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}$$

$$\bar{X} = 0.2$$

$$\bar{Y} = 12,000$$

$$\overline{XY} = 1,640$$

$$\sigma_X = 0.1225$$

$$\sigma_Y = 7,848.57$$

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$$\rho = \frac{1,640 - 0.2 \times 12,000}{0.1225 \times 7,848.57} = -79\%$$

Part b: 1.5 points

Sample Answer 1

$$\tau = 1 - \frac{4Q}{N \times (N - 1)}$$

Reordered		Rank (F)	Rank (S)	Swap 1	Swap 2	Swap 3	Swap 4	Swap 5	Swap 6	Swap 7	Swap 8
Frequency	Severity										
0.05	25,000	1	5	3	3	3	3	3	3	1	1
0.1	12,000	2	3	5	4	4	4	1	1	3	2
0.2	15,000	3	4	4	5	1	1	4	2	2	3
0.25	3,000	4	1	1	1	5	2	2	4	4	4
0.4	5,000	5	2	2	2	2	5	5	5	5	5

$$\begin{array}{ll} Q & 8 \\ N & 5 \\ (N - 1) & 4 \end{array}$$

$$= 1 - \frac{4 \times 8}{5 \times 4} = -60\%$$

Sample Answer 2

Frequency	Severity		
Rank	Rank	Concordant	Discordant
1	5	1	3
2	3		3
3	4	1	1
4	1		1
5	2		
Total		2	8

$$\begin{aligned} \text{Tau} &= (\text{Concordant Pairs} - \text{Discordant Pairs}) / (\text{Concordant Pairs} + \text{Discordant Pairs}) \\ &= (2 - 8) / (2 + 8) = -0.6 \end{aligned}$$

Part c: 1.5 points

Sample Answer 1

It summarizes the dependence in a single number; as such, it doesn't "reveal dependencies that vary along the distributions, such as dependency in one or both tails but not in the middle of the distribution."

Sample Answer 2

It "is heavily influenced by outlying claims with large residuals", because it relates the values of the distributions with weights proportional to the squared residuals.

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Sample Answer 3

It depends on numerical values, not just the relation between the distributions, i.e. it is not an ordinal measure.

Sample Answer 4

It is not invariant under all monotone transformations. Applying certain nonlinear monotone functions to the underlying data will change the correlation statistic.

Part d: 0.5 point

Sample Answer 1

The Kendall statistic also summarizes the dependence in a single number, so it does not address this weakness of the Pearson correlation.

Sample Answer 2

The Kendall statistic weights by the number of swaps (not the square) – as such it isn't as heavily influenced by outlying data points

Sample Answer 3

It is ordinal, as opposed to cardinal; as such it relates the quantiles, not the values, of the distributions

Sample Answer 4

It is invariant under all monotone transformations, as these functions do not change the order of the data points.

Part e: 0.5 point

Advantage Sample Answer 1

Scatterplots show how dependency may change along the distribution, while the two analytical measures calculate dependency over the entire distribution.

Advantage Sample Answer 2

Displays the correlation at different points of the distribution, whereas Pearson's and Kendall's only generate one value for the entire distribution.

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Disadvantage Sample Answer 1

Scatterplots are subjective to the person viewing them – we may imagine relations where none exists, as our eyes see patterns in random dots.

Disadvantage Sample Answer 2

Viewers of scatterplots may confuse patterns from random fluctuation to be a dependency. Example, the ancient Greeks saw constellations among stars in the sky, joining random star positions into figures.

Disadvantage Sample Answer 3

Scatterplots of multivariate distributions portray the marginal distributions along with the dependency among the distributions. One might confuse the shape of marginal distributions with the type of dependency.

EXAMINER'S REPORT

This question requires candidates to calculate and understand basic correlation measures and concepts. Candidates scored very well on this question and generally demonstrated strong command of the material. Common mistakes were largely calculation errors or incomplete written explanations. There were common themes to understand for parts c through e. The majority of candidates demonstrated general understanding of the overall themes/concepts in these parts, whether or not they made a mistake in one or more of the parts.

Part a

Candidates were expected to correctly identify the formula, perform the intermediate calculations and provide the answer. Most mistakes were calculation errors. Some candidates mixed two different calculation methodologies within their response, leading to an incorrect result.

Part b

Candidates were expected to know the formula, understand the method for determining the number of swaps (or concordant/discordant pairs) and correctly perform the calculation. One common mistake was when candidates failed to set up the swap based on the order ranking of either severity or frequency.

Part c

This part requires the candidate to understand the most significant concepts behind the Pearson correlation. They were required to clearly identify two distinct weaknesses of Pearson and briefly describe why these weaknesses occur. The common mistakes were failing to give an incomplete explanation of the weakness and restating one weakness in an alternate manner.

Part d

Candidates were expected to understand the concepts underlying Kendall's Tau and how they compare to the Pearson correlation statistic. In order to receive full credit, candidates needed to identify whether Kendall's Tau addresses the weaknesses identified in part c and give a simple argument why or why not. A common mistake was to give insufficient explanation of why the weakness was or wasn't addressed.

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Part e
Candidates were expected to understand the role of using scatterplots in evaluating correlation. In order to receive full credit, the candidate needed to identify and clearly describe one advantage and one disadvantage to using scatterplots. This part was more difficult. A common error was to state something true about scatterplots that was not clearly a strength or a weakness in evaluating correlations.